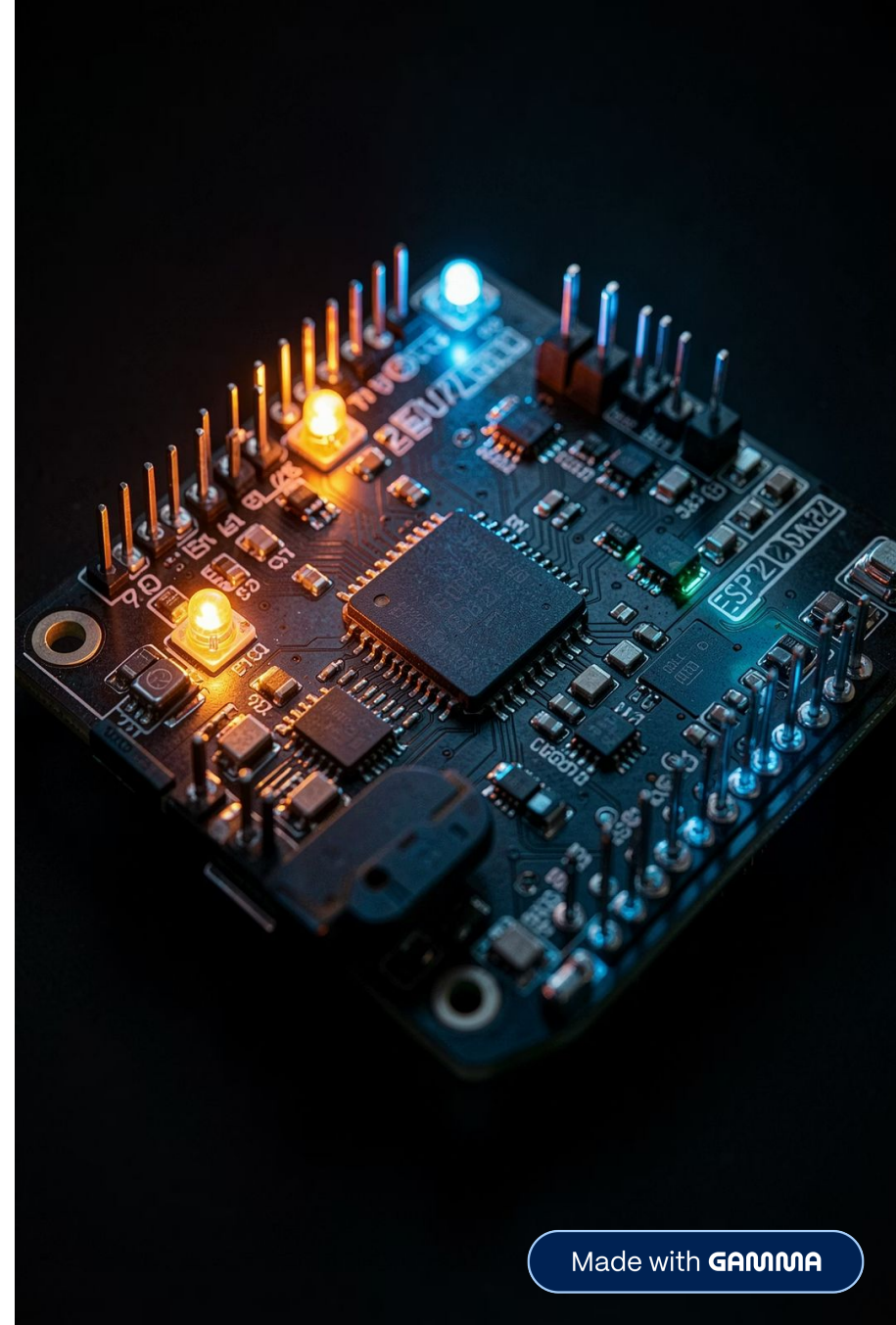


BUILDING CONTROL: ESP32- Based Diorama Control System

A sophisticated embedded firmware platform for intelligent multi-building diorama and model installation control.



Project Overview

An ESP32 firmware system for controlling a multi-building diorama/model installation. Combines WS2811 addressable LED lighting (up to 100 individually addressable driver channels), TMC2209 stepper motor control for mechanical (chain-drive) actuation, Bluetooth Low Energy control from a mobile app, an ESP-NOW wireless mesh so one master unit can command multiple slave ESP32 nodes, and over-the-air firmware updates (both GitHub-hosted and local).



WS2811 LEDs

Up to 100 individually addressable RGB driver channels



TMC2209 Stepper

Chain-drive mechanical actuation with precision control



BLE Control

Mobile app control via Bluetooth Low Energy



ESP-NOW Mesh

Wireless multi-node master/slave coordination

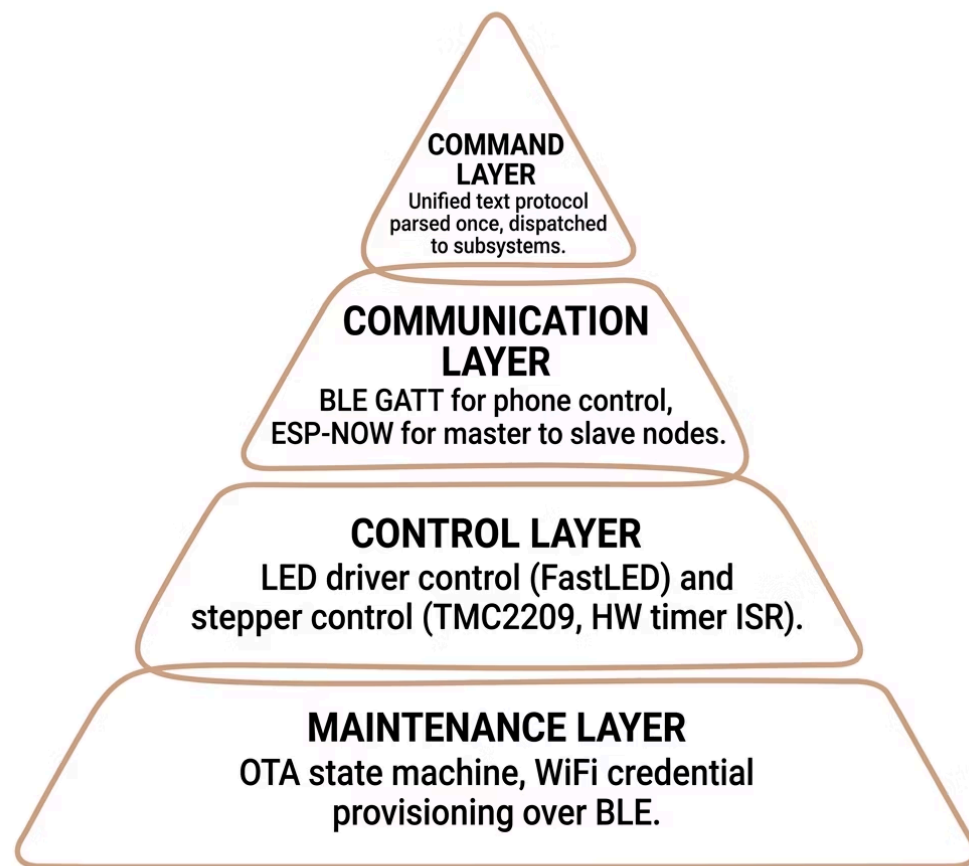


OTA Updates

GitHub-hosted and local over-the-air firmware updates

System Architecture

Layered firmware architecture with clean separation of concerns across four distinct layers:



Command Layer

Unified text-based
command protocol
parsed once,
dispatched to
subsystems

Communication Layer

BLE GATT server
(phone/app control)
plus ESP-NOW
broadcast (master to
slave nodes)

Control Layer

LED driver control
(FastLED/WS2811) and
stepper motor control
(TMC2209 via UART,
hardware timer ISR
stepping)

Maintenance Layer

OTA update state
machine, WiFi credential
provisioning over BLE

Custom Command Protocol

All control flows through one parser: format `#COMMAND:arg1:arg2:arg3:arg4*`

Commands are bounded by `#` and `*`, colon-delimited arguments, up to 4 optional integer arguments.

1	ON / OF Turn LED channel range on/off
2	AN Start animation by ID
3	OA OTA mode select
4	MS Set motor speed and stall threshold
5	MT Motor trigger — direction/steps/current, routable to a specific ESP-NOW node
6	BT Physical button-triggered on/off

This single parser feeds both the BLE characteristic and the ESP-NOW message handler, so the exact same command syntax works locally or when broadcast to another node.

BLE Control Layer



Built on the ESP32 BLEDevice stack with three GATT characteristics under one custom service UUID:

Connection callbacks automatically restart advertising on disconnect, so the device is always discoverable when idle.

Control Characteristic (read/write)

Receives the `#CMD:...*` strings from the mobile app

OTA Characteristic (read/write)

Accepts `"SSID:PASSWORD"` to trigger WiFi-based OTA, and reports back OTA status: `NoOTA`, `OTAGit`, `OTAInProgress`, `OTACompleted`, `OTAFailed`

Notify Characteristic

Reserved for pushing status back to the app



LED Driver Control (WS2811 / FastLED)

Up to 100 addressable RGB "driver" channels represent building windows/lights, controlled via the FastLED library.

Individual channel or channel-range ON/OFF addressing (byte-level R/G/B addressing across the whole array).



Instant On/Off Patterns

Immediate channel switching for crisp, responsive lighting control



Sequential RGB Fade

Channel-by-channel fade in/out across the full RGB spectrum



RGB Crossfade

Smooth color transitions blending between RGB values

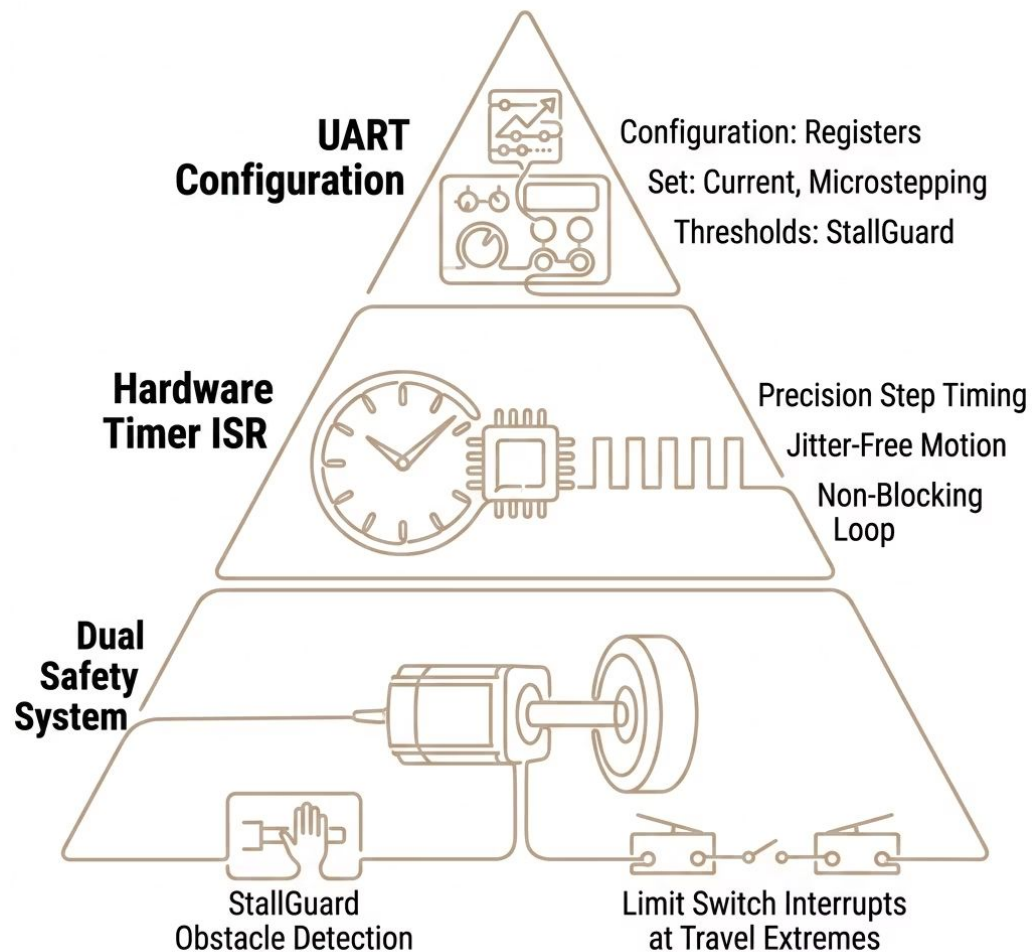


Flash-then-Fade

Dramatic flash burst followed by a smooth fade-out sequence

All six built-in animation routines are interruptible mid-animation by any incoming command via a shared `CHECK_INTERRUPT` macro, so a new command always takes priority over a running animation.

Stepper Motor Control (TMC2209)



Chain-drive / mechanical actuation using a TMC2209 stepper driver over UART (register configuration: current, microstepping, StallGuard threshold).

Hardware Timer ISR

Precision step timing handled entirely in a hardware timer ISR (not blocking loop() code) for jitter-free motion.

Dual Safety Layer

StallGuard-based obstacle/end-stop detection, backed up by physical limit switch interrupts at both travel extremes for hard-stop safety.

Runtime Adjustable

Speed (step delay), motor current, and direction — all driven from the same command protocol.

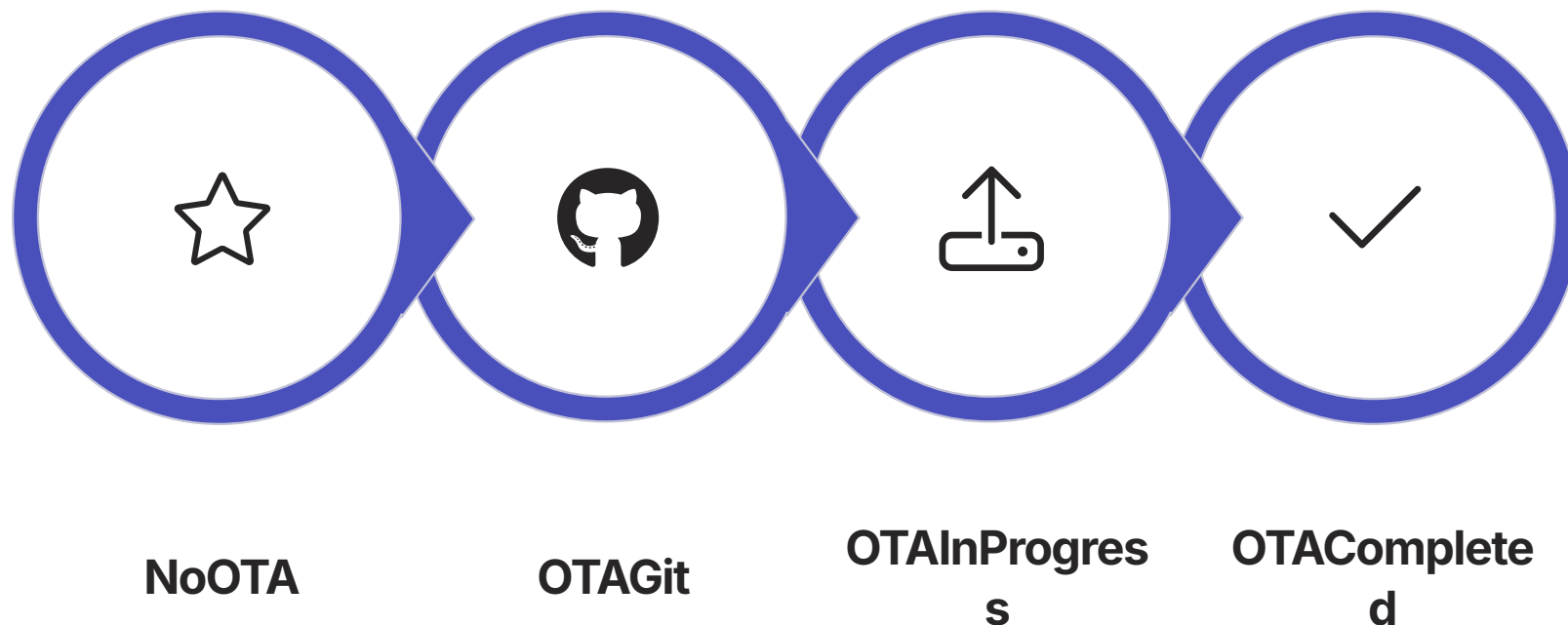
ESP-NOW Wireless Mesh & OTA Update System

ESP-NOW Mesh

Connectionless protocol links multiple ESP32 nodes without a WiFi access point. One master node (ID 0) can broadcast a motor command to all slaves, or target a specific node by ID — enabling synchronized or independent control of multiple physical units from a single BLE connection to the master. Commands received over ESP-NOW use the identical parser used for BLE.

OTA Updates

Two paths — GitHub-hosted binary OTA (fetches firmware over HTTPS, flashes via the ESP32 Update library, with progress tracking and timeout handling) and local network OTA (ArduinoOTA for fast dev iteration). A status state machine tracks progress: NoOTA → OTAGit → OTAInProgress → OTACompleted or OTAFailed. WiFi credentials can be provisioned live over BLE.



The state machine ensures robust tracking of every firmware update attempt, from idle through completion or failure, with full status reporting back to the mobile app over BLE.

Design Highlights



Unified Command Parser

Single parser shared across BLE and ESP-NOW transports — one syntax, two transports



Non-Blocking Motor Control

Hardware timer interrupts keep LED animation and BLE handling responsive during motion



Dual Safety Layer

Software StallGuard plus physical limit switch interrupts for motor limits



Interruptible Animations

Any new command instantly pre-empts a running LED sequence



Remote OTA Updates

BLE-provisioned WiFi and GitHub-hosted OTA — no physical access required

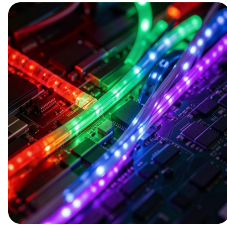
Tech Stack

ESP32 (WROOM / S3 variants supported via conditional compilation), Arduino framework, FastLED, TMCStepper library, ESP32 BLE stack, ESP-NOW, ArduinoOTA / HTTPClient + Update library, custom lightweight text command protocol.



ESP32 Platform

WROOM / S3 variants via conditional compilation



FastLED

WS2811 addressable LED driver library



TMCStepper

TMC2209 UART stepper motor control library



ESP32 BLE + ESP-NOW

BLE GATT stack and connectionless wireless mesh



OTA Libraries

ArduinoOTA, HTTPClient, and ESP32 Update library